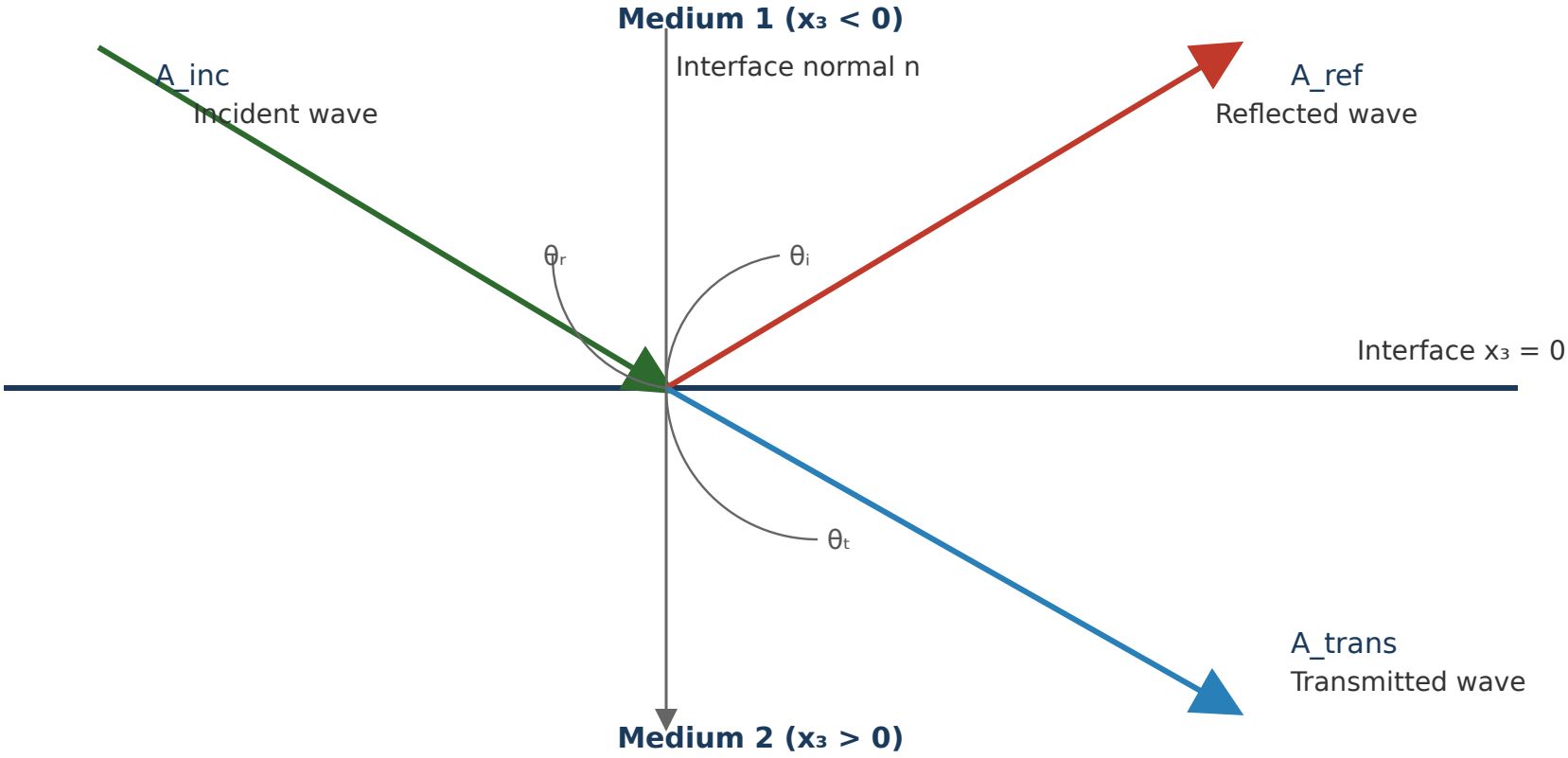
Interface reflection and transmission

Incident, reflected, and transmitted waves at a bonded interface between two media



Interface conditions (perfect bonding):

$u^{(1)}_i = u^{(2)}_i$ (displacement continuity)

$\sigma^{(1)}_{ji} n_j = \sigma^{(2)}_{ji} n_j$ (traction equilibrium)

$T^{(1)} = T^{(2)}$ (temperature continuity)

$q^{(1)}_i n_i = q^{(2)}_i n_i$ (heat flux continuity)

$\phi^{(1)} = \phi^{(2)}$, $D^{(1)}_n = D^{(2)}_n$ (electrical continuity)

Interface-plane accounting:

$R\Sigma + T\Sigma + A_{int} = 1$

$R\Sigma$: reflected normal-flux ratio at interface

$T\Sigma$: transmitted normal-flux ratio at interface

A_{int}: local interface residual (not bulk loss)

all modes and cross terms retained